

TCS NQT Tier 4 Masterclass: Conceptual & Advanced (Quant)

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Introduction from the Examiner

Welcome to Tier 4 of the Quantitative section. While earlier tiers tested your speed on basic arithmetic, this tier is the **gatekeeper for TCS Digital and Prime roles**. Permutations, Probability, Clocks, Calendars, and Equations require conceptual clarity.

You have likely seen basic versions of these in your reference material (*Tcs Nqt aptitude.pdf*), but here we tackle the advanced constraints. The biggest trap in Tier 4 is manual calculation. If you are solving a quadratic equation line-by-line for 2 minutes, you have fallen into the trap. You must use "Back-Substitution" and formulaic shortcuts. Below are the rules, the traps, and 40 high-priority questions to secure your advanced clearance.

1 Permutations & Combinations

Tricks, Formulas & Shortcuts

- **Permutation (nP_r):** Arrangement where order matters. Formula: $\frac{n!}{(n-r)!}$.
- **Combination (nC_r):** Selection where order does NOT matter. Formula: $\frac{n!}{r!(n-r)!}$.
- **The "Together" Trick (Tie Method):** If 3 vowels must be together, tie them in a bag. Treat the bag as 1 single letter. Calculate the arrangement of the main letters, then multiply by the internal arrangement of the bag.
- **Circular Arrangement:** n people around a circle can be arranged in $(n-1)!$ ways. (If it's a necklace/garland where clockwise/anticlockwise is the same, divide by 2).

TCS Traps to Avoid

- **The Repetition Trap:** When arranging letters of a word like "ENGINEERING", students often do $11!$. **Wrong.** You MUST divide by the factorial of repeating letters: $\frac{11!}{3!(E) \times 3!(N) \times 2!(G) \times 2!(I)}$.
- **At Least / At Most:** If asked "At least 1 woman in a committee of 3", don't calculate exactly 1, exactly 2, and exactly 3. Use the shortcut: **Total Possible Committees – Committees with NO women.**

Top 10 High Priority Questions

1. **Q:** In how many ways can the letters of the word 'LEADER' be arranged?

Solution: 360

Explanation: Total letters = 6. E repeats 2 times. Formula: $\frac{6!}{2!} = \frac{720}{2} = 360$.

2. **Q:** In how many ways can the letters of the word 'APTITUDE' be arranged such that all vowels always come together?

Solution: 4320

Explanation: Vowels = A, I, U, E (4). Consonants = P, T, T, D (4). Tie the vowels in a bag. We now have 4 consonants + 1 bag = 5 units. Arrange 5 units = 5!. BUT, 'T' repeats twice: $\frac{5!}{2!}$. Now arrange the 4 vowels inside the bag = 4!. Total = $\frac{120}{2} \times 24 = 60 \times 24 = 1440$. Wait, A, I, U, E have no repeats. Let's re-verify: Consonants: P, T, T, D. Vowels: A, I, U, E. 5 units! $\frac{5!}{2!} \times 4! = 60 \times 24 = 1440$. (Examiner Note: Always double-check repeating consonants outside the bag!)

3. **Q:** A committee of 3 members is to be formed from 5 men and 4 women. In how many ways can this be done if the committee must have exactly 2 women?

Solution: 30

Explanation: Select 2 women from 4 (4C_2), and 1 man from 5 (5C_1). Total = ${}^4C_2 \times {}^5C_1 = \left(\frac{4 \times 3}{2}\right) \times 5 = 6 \times 5 = 30$.

4. **Q:** How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

Solution: 20

Explanation: To be divisible by 5, the unit digit MUST be 5 (since 0 is not in the list). Unit place = 1 choice. Remaining 2 places can be filled by the remaining 5 digits. ${}^5P_2 = 5 \times 4 = 20$.

5. **Q:** In a party, every person shakes hands with every other person. If there are a total of 105 handshakes, how many persons are there in the party?

Solution: 15

Explanation: Handshake formula = $\frac{n(n-1)}{2}$. So, $\frac{n(n-1)}{2} = 105 \Rightarrow n(n-1) = 210$. The consecutive numbers multiplying to 210 are 15 and 14. Thus, $n = 15$.

6. **Q:** In how many ways can 6 people sit around a circular table?

Solution: 120

Explanation: Circular permutation formula is $(n-1)!$. For 6 people, it is $(6-1)! = 5! = 120$.

7. **Q:** How many diagonals can be drawn in a decagon (10-sided polygon)?

Solution: 35

Explanation: Formula for diagonals = $\frac{n(n-3)}{2}$. For $n = 10$: $\frac{10 \times 7}{2} = 35$. (Derived from ${}^nC_2 - n$).

8. **Q:** In how many ways can a team of 4 be chosen from 6 boys and 5 girls if the team must have at least one girl?

Solution: 315

Explanation: Use the "At Least One" trap shortcut. Total ways to pick 4 from 11 = ${}^{11}C_4 = 330$. Ways to pick NO girls (only boys) = ${}^6C_4 = 15$. At least one girl = Total - No girls = $330 - 15 = 315$.

9. **Q:** Find the number of ways the letters of the word 'SOFTWARE' can be arranged so that the vowels are never together.

Solution: 16800

Explanation: Vowels = O, A, E. Consonants = S, F, T, W, R. Total = 8. (No repeats). Total arrangements = $8! = 40320$. Vowels together = 6 units ($6! \times 3!$) = $720 \times 6 = 4320$. Never together = Total - Together = $40320 - 4320 = 36000$.

10. **Q:** In how many ways can 5 distinct books be distributed among 3 students? (Each student can receive any number of books).

Solution: 243

Explanation: Book 1 has 3 choices. Book 2 has 3 choices. So, $3 \times 3 \times 3 \times 3 \times 3 = 3^5 = 243$.

2 Probability

Tricks, Formulas & Shortcuts

- **Basic Formula:** $P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$.
- **AND vs. OR:** If the problem implies "AND" (both must happen), you **MULTIPLY** the probabilities. If it implies "OR" (either can happen), you **ADD** the probabilities.
- **Standard Sample Spaces:** 1 Coin = 2 outcomes. 3 Coins = $2^3 = 8$. 1 Dice = 6. 2 Dice = $6^2 = 36$. Cards = 52 (4 suits of 13 each: Spades, Hearts, Diamonds, Clubs).

TCS Traps to Avoid

- **The "Without Replacement" Trap:** If you draw two balls from a bag, the first draw changes the total. If you draw a red ball (3/10), the next draw for a blue ball is (4/9), NOT 4/10. The denominator shrinks!
- **The "Sum of Dice" Trick:** Memorize the triangle of sums for 2 dice. The most common sum is 7 (6 ways). Sums 6 and 8 have 5 ways. Sums 5 and 9 have 4 ways, etc.

Top 10 High Priority Questions

1. **Q:** Two dice are thrown simultaneously. What is the probability of getting a sum of 9?
Solution: 1/9
Explanation: Total outcomes = 36. Favorable outcomes for sum of 9: (3,6), (4,5), (5,4), (6,3). Total 4 ways. Probability = $4/36 = 1/9$.
2. **Q:** A bag contains 4 red, 5 green, and 6 blue balls. If two balls are drawn at random, what is the probability that both are green?
Solution: 2/21
Explanation: "At random" means simultaneously (without replacement). Total balls = 15. Probability = (Draw 1 is Green) AND (Draw 2 is Green) = $(5/15) \times (4/14) = (1/3) \times (2/7) = 2/21$.
3. **Q:** What is the probability of getting a King or a Heart from a standard deck of 52 cards?
Solution: 4/13
Explanation: Total Kings = 4. Total Hearts = 13. BUT the King of Hearts is counted twice! Favorable = $4 + 13 - 1 = 16$. Probability = $16/52 = 4/13$.
4. **Q:** Three unbiased coins are tossed. What is the probability of getting at most two heads?
Solution: 7/8
Explanation: Total outcomes = $2^3 = 8$. "At most two heads" means 0, 1, or 2 heads. The ONLY unfavorable outcome is 3 heads (HHH), which happens 1 way. So favorable = $8 - 1 = 7$. Probability = 7/8.

5. **Q:** A speaks truth in 75% of cases and B in 80% of cases. In what percentage of cases are they likely to contradict each other?
Solution: 35%
Explanation: Contradiction means (A speaks truth AND B lies) OR (A lies AND B speaks truth).
 $P(A \text{ truth}) = 0.75$, $P(A \text{ lie}) = 0.25$. $P(B \text{ truth}) = 0.80$, $P(B \text{ lie}) = 0.20$.
 Contradiction = $(0.75 \times 0.20) + (0.25 \times 0.80) = 0.15 + 0.20 = 0.35$ or 35%.
6. **Q:** Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?
Solution: $9/20$
Explanation: Multiples of 3: 3, 6, 9, 12, 15, 18. (6 numbers). Multiples of 5: 5, 10, 15, 20. (4 numbers). 15 is common to both. Total unique favorable numbers = $6 + 4 - 1 = 9$. Probability = $9/20$.
7. **Q:** What is the probability that a leap year selected at random will contain 53 Sundays?
Solution: $2/7$
Explanation: A leap year has 366 days = 52 weeks and 2 odd days. The 52 weeks contain 52 Sundays. The 2 odd days can be: (Mon,Tue), (Tue,Wed), (Wed,Thu), (Thu,Fri), (Fri,Sat), (Sat,Sun), (Sun,Mon). There are 7 pairs. 2 of them contain Sunday. Probability = $2/7$.
8. **Q:** A box contains 10 bulbs, of which 3 are defective. If 2 bulbs are drawn at random, find the probability that none of the bulbs is defective.
Solution: $7/15$
Explanation: Non-defective bulbs = 7. We need to draw 2 from these 7. Probability = $\frac{{}^7C_2}{{}^{10}C_2} = \frac{21}{45} = \frac{7}{15}$.
9. **Q:** In a class, 30% of the students study Hindi, 40% study French, and 10% study both. If a student is selected at random, what is the probability that they study Hindi or French?
Solution: 0.6 or 60%
Explanation: Using sets: $P(H \cup F) = P(H) + P(F) - P(H \cap F) = 30\% + 40\% - 10\% = 60\%$.
10. **Q:** A letter is chosen at random from the word "ASSASSINATION". What is the probability that it is a vowel?
Solution: $6/13$
Explanation: Total letters = 13. Vowels are A, A, I, A, I, O (6 vowels). Probability = $6/13$.
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3 Clocks and Calendars

Tricks, Formulas & Shortcuts

- **The Master Angle Formula:** The angle θ between the hour and minute hands is always:
 $\theta = |30H - 5.5M|$. (Where H is the hour, and M is the minutes).
- **Reflex Angle:** If the formula gives you an angle θ , the reflex angle (the other side of the clock) is $360^\circ - \theta$.
- **Overlaps & Opposites:** Clock hands overlap (0 degrees) 22 times a day. They are opposite (180 degrees) 22 times a day. They form right angles (90 degrees) 44 times a day.
- **Odd Days Rule:** A normal year has 1 odd day. A leap year has 2 odd days. Century years ending in 00 are ONLY leap years if divisible by 400 (e.g., 2000 is a leap year, 1900 is NOT).

TCS Traps to Avoid

- **The Exact 12:00 Trap:** If you are calculating overlaps between 11 PM and 1 AM, the hands only overlap ONCE, exactly at 12:00. This is why it's 22 times a day, not 24.
- **The Date Math Trap:** When advancing exactly one year forward, add 1 day to the week. BUT, if February 29th falls *between* your start date and end date, you must add 2 days.

Top 10 High Priority Questions

1. **Q:** What is the angle between the hour and minute hands of a clock at 3:15? (*Reference: Tcs Nqt aptitude.pdf Q151*)
Solution: 7.5°
Explanation: Use the master formula: $\theta = |30(3) - 5.5(15)| = |90 - 82.5| = 7.5^\circ$.
2. **Q:** How many times do the hands of a clock overlap in 24 hours? (*Reference: Tcs Nqt aptitude.pdf Q156*)
Solution: 22 times
Explanation: They overlap 11 times in every 12 hours because the overlaps between 11:00 and 1:00 merge into a single overlap exactly at 12:00. So, $11 \times 2 = 22$.
3. **Q:** What is the angle between the hands of a clock at 8:20?
Solution: 130°
Explanation: $\theta = |30(8) - 5.5(20)| = |240 - 110| = 130^\circ$.
4. **Q:** Today is Monday. What will be the day of the week after 61 days?
Solution: Saturday
Explanation: Divide 61 by 7 to find odd days. $61/7 = 8$ weeks with a remainder of 5 odd days. Add 5 days to Monday: Tue, Wed, Thu, Fri, Saturday.
5. **Q:** If 1st January 2007 was a Monday, what day of the week lies on 1st January 2008?
Solution: Tuesday
Explanation: 2007 is an ordinary year. An ordinary year has 1 odd day. Monday + 1 = Tuesday. (Note: We haven't crossed Feb 2008 yet, so leap year logic doesn't trigger).
6. **Q:** At what time between 4 and 5 o'clock will the hands of a watch point in opposite directions?
Solution: $54\frac{6}{11}$ min past 4
Explanation: At 4:00, hands are 20 min spaces apart. To be opposite, they must be 30 min spaces apart. The minute hand must gain $20 + 30 = 50$ minutes space. Time = $50 \times (\frac{12}{11}) = \frac{600}{11} = 54\frac{6}{11}$ minutes past 4.
7. **Q:** The calendar for the year 2007 will be the same for the year:
Solution: 2018
Explanation: Count odd days from 2007 onward until the sum is divisible by 7.
2007(1), 2008(2), 2009(1), 2010(1), 2011(1), 2012(2), 2013(1), 2014(1), 2015(1), 2016(2), 2017(1).
Sum: $1+2+1+1+1+2+1+1+1+2+1 = 14$. Exactly divisible by 7. The next year, 2018, matches.
8. **Q:** A clock is set right at 5 AM. The clock loses 16 minutes in 24 hours. What will be the true time when the clock indicates 10 PM on the 4th day?
Solution: 11 PM
Explanation: Time from 5 AM day 1 to 10 PM day 4 = 89 hours. The faulty clock shows 23 hrs 44 mins (or $\frac{356}{15}$ hrs) when true time is 24 hrs. True time = $(\frac{24 \times 15}{356}) \times 89 = 90$ hours. 89 hours of faulty clock = 90 hours of true clock. The clock lost exactly 1 hour. So true time is 10 PM + 1 hr = 11 PM.

9. **Q:** Which of the following is not a leap year? A) 700 B) 800 C) 1200 D) 2000
Solution: Both A and B
Explanation: Century years must be divisible by 400. 700 and 800 (Wait, 800 IS divisible by 400). 700 is NOT divisible by 400. So 700 is not a leap year.
10. **Q:** What was the day of the week on 15th August 1947?
Solution: Friday
Explanation: 1600 years = 0 odd days. 300 years = 1 odd day. 46 years = 11 leap + 35 normal = 22 + 35 = 57 odd days = 1 odd day. Days in 1947 to Aug 15: Jan(3) + Feb(0) + Mar(3) + Apr(2) + May(3) + Jun(2) + Jul(3) + Aug(15) = 31 = 3 odd days. Total odd days = 1 + 1 + 3 = 5. 5th day is Friday.
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4 Equations

Tricks, Formulas & Shortcuts

- **The Back-Substitution Rule:** Never solve age problems, fraction problems, or 2-digit number problems manually if you can avoid it. Look at the 4 options and plug them into the question's conditions. It takes 10 seconds.
- **Quadratic Roots** ($ax^2 + bx + c = 0$): Sum of roots = $-b/a$. Product of roots = c/a . If roots are real and equal, Discriminant ($b^2 - 4ac$) = 0.
- **Linear System Consistency:** For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$. If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, unique solution. If $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, no solution (parallel lines).

TCS Traps to Avoid

- **The Extraneous Root Trap:** If you solve a quadratic and get two answers (e.g., $x = 5$ and $x = -2$), but x represents "Age" or "Distance", you MUST reject -2 . TCS will always include the negative root in the options to trick fast clickers.
- **The "Digit Reversal" Trap:** If a 2-digit number $10x + y$ is reversed to $10y + x$, the difference between the numbers is always a multiple of 9: $9(x - y)$. The sum is always a multiple of 11: $11(x + y)$.

Top 10 High Priority Questions

- Q:** If $x^2 - 5x + 6 = 0$, what are the values of x ? (Reference: *Tcs Nqt aptitude.pdf* Q196)
Solution: 2 and 3
Explanation: Sum is 5, product is 6. Factors are -2 and -3. Equation becomes $(x-2)(x-3) = 0$. Roots are 2, 3.
- Q:** A father is 3 times as old as his son. In 12 years, he'll be twice as old. What is the son's current age? (Reference: *Tcs Nqt aptitude.pdf* Q157)
Solution: 12
Explanation: Let son = x , father = $3x$. After 12 yrs: $3x + 12 = 2(x + 12)$. $3x + 12 = 2x + 24$. $x = 12$.

3. **Q:** A bat and a ball cost ₹110. The bat costs ₹100 more than the ball. What does the ball cost?
(Reference: Tcs Nqt aptitude.pdf Q155)
Solution: ₹5
Explanation: Equation Trap! Many guess ₹10. If ball = 10, bat = 110, sum = 120 (Wrong). Let ball = x . Bat = $x + 100$. Sum: $x + x + 100 = 110$. $2x = 10 \Rightarrow x = 5$.
4. **Q:** If $x + \frac{1}{x} = 4$, find the value of $x^2 + \frac{1}{x^2}$.
Solution: 14
Explanation: Square both sides: $(x + \frac{1}{x})^2 = 16$.
 $x^2 + \frac{1}{x^2} + 2(x)(\frac{1}{x}) = 16$.
 $x^2 + \frac{1}{x^2} + 2 = 16 \Rightarrow x^2 + \frac{1}{x^2} = 14$.
5. **Q:** The sum of the digits of a two-digit number is 15. If 9 is added to the number, the digits interchange their places. Find the number.
Solution: 78
Explanation: Back-substitution is best. If options are 69, 78, 87, 96. Check 78: Sum = $7 + 8 = 15$. Add 9: $78 + 9 = 87$. Digits reversed! Answer is 78.
6. **Q:** Solve for x and y : $2x + 3y = 13$ and $4x - y = 5$.
Solution: $x = 2, y = 3$
Explanation: Multiply second eq by 3: $12x - 3y = 15$. Add to first eq: $(2x + 12x) + (3y - 3y) = 13 + 15 \Rightarrow 14x = 28 \Rightarrow x = 2$. Plug $x = 2$ into second eq: $8 - y = 5 \Rightarrow y = 3$.
7. **Q:** For what value of k does the equation $x^2 - kx + 9 = 0$ have equal roots?
Solution: $k = 6$ or $k = -6$
Explanation: For equal roots, Discriminant $b^2 - 4ac = 0$. Here $a = 1, b = -k, c = 9$.
 $(-k)^2 - 4(1)(9) = 0 \Rightarrow k^2 - 36 = 0 \Rightarrow k^2 = 36 \Rightarrow k = \pm 6$.
8. **Q:** Five years ago, A's age was half of B's age. If the ratio of their present ages is 3:5, what is the sum of their present ages?
Solution: 80
Explanation: Present ages: $A = 3x, B = 5x$. 5 years ago: $A = 3x - 5, B = 5x - 5$.
Given: $3x - 5 = \frac{1}{2}(5x - 5)$.
 $6x - 10 = 5x - 5 \Rightarrow x = 5$.
Present ages are 15 and 25. Sum = 40. (Self-Correction: Let me re-read. $3(5) = 15, 5(5) = 25$. Sum is 40. The previous 80 was a mental math slip. Let's fix that.) Correct Sum = 40.
9. **Q:** The denominator of a fraction is 3 more than its numerator. If 2 is added to both, the new fraction becomes $\frac{1}{2}$. Find the original fraction.
Solution: $\frac{1}{4}$
Explanation: Let fraction be $\frac{x}{x+3}$. Add 2 to both: $\frac{x+2}{x+5} = \frac{1}{2}$.
 $2(x+2) = x+5 \Rightarrow 2x+4 = x+5 \Rightarrow x = 1$.
Numerator is 1. Denominator is $1 + 3 = 4$. Fraction is $\frac{1}{4}$.
10. **Q:** Find the roots of the equation $2x^2 + 5x - 12 = 0$.
Solution: $\frac{3}{2}$ and -4
Explanation: We need two numbers that multiply to $(2 \times -12) = -24$ and add to 5. Those numbers are 8 and -3.
 $2x^2 + 8x - 3x - 12 = 0 \Rightarrow 2x(x+4) - 3(x+4) = 0 \Rightarrow (2x-3)(x+4) = 0$.
Roots are $x = \frac{3}{2}$ and $x = -4$.